multigrain

_travelers guide
_for navigating particle storms

8 sounds	2 scenes per sound	32 s per sample

[rec] press [rec] hold › then 1 s = keep holding [load] grey = alt label

home page unless marked · alt › = press alt first · knobs edit the blinking sound + scene

samples = .wav · 48_khz · 16-bit · stereo · ≤ 32_sec

red = hold

01 sounds — mgr_01

[1–8]	press / hold = play grains · active sound blinks
[latch] › [1–8]	**latch** it on · switching keeps it latched
[latch] + [1–8]	select silently, without playing
[1–8] + [1–8]	**copy sound** to an empty sound
[latch] + [clear]	**kill all grains** (blur tails on)
[gate] [next]	play the sound · step one right (normalled)
[select]	0–5 v picks 1–8 · empties skipped

02 scenes · morph — mgr_02

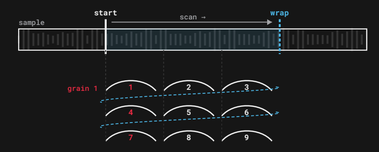
[scene a] [b]	which scene the knobs edit (blinks)
fader	a ← → b · you always hear the fader
[scene] + [scene] / [1–8]	**copy scene** to the other one or a sound
[rand] / [clear] + [scene]	randomize / reset that scene
[morph] cv	±5 v = full sweep · knob attenuverts
shape · reverse	don't blend: each grain picks a or b by chance

03 position — mgr_03

[start]	where grains begin in the sample
[wrap]	how far scan travels before wrapping
[scan]	noon = still · cw forward · ccw back
resets	sound press or gate = back to start
live sounds	start cw = the newest audio coming in
[reverse] long	**ping-pong**: bounce, don't wrap (fades)

__fig.01 : the scan

start · wrap · scan



reaching wrap sends the next grain back to start

noon = every grain from start · ccw scans backward · ping-pong bounces

04 grain — mgr_04

[rate]	0 hz (one grain per trigger) – 130.8 hz
[size]	20 ms – 4 s · noon ≈ 1 s
[pitch]	±2 oct (±3 with mod) · green = octaves
[tone]	ccw lowpass · noon open · cw highpass
[level] [blur]	grain volume · reverb send + decay
[reverse]	grains play backwards (per scene)



tukey = default · discrete: morph picks per grain by chance

05 links · sync — mgr_05

[rate–size]	size sets the rate · rate = 1× – 8×
[size–pitch]	size = audio amount: octave up = half
alt › [sync]	rate ×/÷ 2 4 8 16 of clock · noon ×1
sync + [rate–size]	size snaps to clock divisions



rate–size linked: size sets the base rate, rate multiplies it

07 clear · control all — mgr_07

[clear] › item ×2	**safe clear** · red = clearable
[clear] + item	quick clear, no confirm
items	sound (×1 settings, ×2 sample) · scene · mod · sample memory · load = init · quant
[scene] + turn	value goes to both scenes
[latch] + turn	every sound · + scene = everything
[rand] + [1–8]	randomize a whole sound

12 quantizer — mgr_12

alt › [quant] / [quant]	open it (hold quant from home)
keys	clear rand x y z black · 1–7 white
notes	root yellow · scale purple · hold = root
[quant]	knob + mod (dim) / mod only (lit)
[latch] + note	apply to every sound

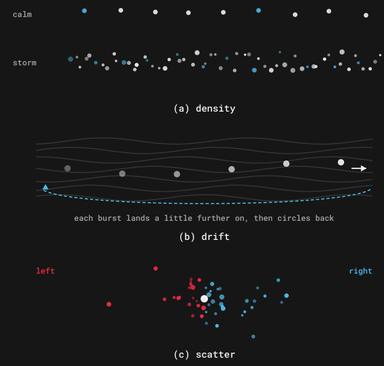
06 modulation — mgr_06

[rand] [x] [y] [z]	mod page · hold = momentary
mod page	knobs attenuvert: cw + yellow, ccw – purple
[rand] / [x] + turn	quick assign from home
[x] [y] [z] jacks	±5 v = full range · pitch 100% = 1v/oct
[rand] + [x] / [alt]	randomize x's (or rand's) assigns
[clear] › turn	on a mod page: clear that knob
rand · start	cw spray · ccw = anywhere in wrap
rand · rate	ccw = trigger chance 100% → 3%
rand · level, tone	ccw = random stereo (l / r opposite)

13 sample loader — mgr_13

alt › [/project] [/wavs] [/recs]	load from that folder
[‹prev] [next›]	step · hold = 5 at a time
[1–8]	preview through the sound's grains

__fig.02 : the storm — mm_04



08 sampling — mgr_08

[sample]	rec (red) · again = stop · 32 s max
then	blue sample = save · empty 1–8 = assign
[sample] 1 s	grab the last 32 s · or open trim
trim page	start / end · scene a / b preview · dir · scan = nudge 25–250 ms
[alt] + [sample]	settings: gain 0.5–8.5× · thrsh · norm
[thrsh]	rec waits for the input (sample fades red)
[resamp]	inputs · resample outs · looping (magenta)

09 live · looping — mgr_09

[1–8] + [sample]	**live sound** from the inputs (magenta)
[latch] + [1–8] + [sample]	same, no trigger
[sample]	magenta = recording · cyan = frozen
[sample] 1 s	save the loop · 8–32 s memory
[thrsh]	auto-freezes when the input goes quiet

10 alt page — mgr_10

[alt] / [alt]	grey labels · hold = momentary
alt › [pitch]	fine-tune ±7 semitones
alt › [a] [b]	cycle output modes (see colours)
[1–8] on alt	sample live · load choke · save gate
[1–8] + [save] 1 s	save default sound (load = reset)

11 presets · projects — mgr_11

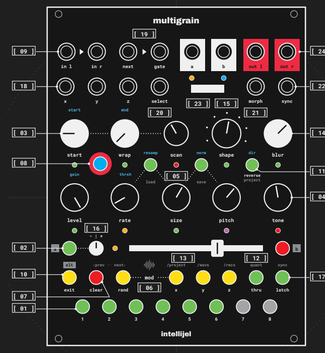
alt › [load] [save] [project]	preset / project pages
[x] [y] [z] › [1–8]	bank (twice = 4–6) › slot
[save] / [load]	quick save · quick reload
[1–8] 4 s	lock preset · collect project
[clear] + [load]	fresh empty preset
room	48 projects × 48 presets × 8 sounds

__fig.03 : front panel

01–13 = sections · 14–24 = hardware

[01] _1–8 : the eight sounds
[02] _scene a / b : scenes, fader
[03] _start wrap scan : position
[04] _grain knobs : six grey knobs
[05] _links : rate–size, size–pitch
[06] _mod : rand · x · y · z
[07] _clear : clear · reset
[08] _sample : record · freeze
[09] _in l / r : audio in · live
[10] _alt / exit : alt page
[11] _reverse : reverse · project
[12] _quant : quantizer
[13] _alt › x y z : sample loader

[14] _blur : reverb
[15] _shape : 7 grain windows
[16] _morph att : cv attenuverter
[17] _thru / latch : monitor · latch
[18] _x y z : cv in ±5 v
[19] _next / gate : step · play
[20] _select : 0–5 v sound pick
[21] _morph : fader cv ±5 v
[22] _sync : clock in
[23] _a / b : cv outs + microsd
[24] _out l / r : audio out



_where things hide

25 · mgr_25

system settings

alt › hold project 4 s

- **rand** random: unipolar (default) / bipolar
- **latch** next jack = scan reset
- **x** auto-collect on preset save
- **start** gate delay, up to 10 ms
- **wrap** select span 5 v › 0.667 v (keyboard)
- **size** max grain: 4 s · ≤ 4 s / sample · ≤ 8 s · full
- **rate–size** shrink grains at high rates
- **clear** 30 s idle = back home (on)
- **thru** knobs jump (default) / catch
- **pitch** a / b 1v/oct trim · size–pitch = test notes

button LEDs

- **lit** on / available
- **dim** off, but can be turned on
- **off** not available here
- **blinking** selected (active sound / scene)
- **rapid** that page is open
- **fading** leads to a page or an action

microsd · boot

- **card** fat32 · 32 gb card included
- **folders** Project01–48 · /Wavs · /Recs
- **latch + clear** hold 4 s = reboot
- **at boot** clear = reset globals · sample / load / save / project = LED theme

LED colours

default theme · no screen, so colour is the display

1–8	loaded · playing · missing · live · live playing
sample	listening · recording · looping · frozen
mod · pitch	+ mod · – mod · quantized · octave · between
a / b outs	morph · grain trig · grain env · in env · out env · random · pitch · rec · in gate
firmware	version, shown at boot

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · colors from the intellijel multigrain manual
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